B3.1 Gas exchange (HL)

NAME	DATE	TOTAL
		_____ / 30

About 30 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 DEFINE cooperative binding. [B3.1.13] [1]

2 STATE the largest number of oxygen molecules one haemoglobin molecule can carry. [B3.1.11] [1]

3 STATE the direction in which a rise in CO_2 shifts the oxygen dissociation curve. [B3.1.12] [1]

4 STATE the direction in which the curve for foetal haemoglobin lies compared with adult haemoglobin. [B3.1.11] [1]

5 STATE a typical partial pressure of oxygen in the alveoli. [B3.1.13] [1]

6 OUTLINE what is meant by the partial pressure of oxygen. [B3.1.13] [2]

Section B: Understanding

12 MARKS

7 EXPLAIN why the oxygen dissociation curve of haemoglobin is S-shaped. [B3.1.13] [3]

8 EXPLAIN why foetal haemoglobin must have a higher affinity for oxygen than adult haemoglobin. [B3.1.11] [3]

9 **EXPLAIN** how a rise in carbon dioxide causes haemoglobin to release oxygen. [B3.1.12] [3]

10 **EXPLAIN** how the Bohr shift benefits a muscle during hard exercise. [B3.1.12] [3]

Section C: Working with evidence

7 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [B3.1.11, B3.1.13]

Llamas live high in the Andes, where the air in the alveoli has a pO_2 of about 6 kPa. The table shows the saturation of llama and human haemoglobin at different partial pressures of oxygen.

PO_2 / KPA	1	2	4	6	8	13
HUMAN SATURATION / %	3	18	59	81	90	97
LLAMA SATURATION / %	7	33	76	91	95	99

a **COMPARE** the saturation of the two haemoglobins in the lungs of an animal at high altitude. [1]

b **CALCULATE** the fall in saturation of each haemoglobin as blood passes from lungs at 6 kPa to active muscle at 2 kPa. [2]

c **EXPLAIN** the advantage of llama haemoglobin at high altitude. [2]

d **DEDUCE** where the llama curve lies compared with the human curve. [1]

e **SUGGEST** one disadvantage of a haemoglobin with a high affinity for oxygen. [1]

Section D: Longer answer

4 MARKS

12 **COMPARE** the effect of foetal haemoglobin and the effect of high CO_2 on the oxygen dissociation curve. [4]

[B3.1.11, B3.1.12]